BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022 • Paper Code: Biology Paper–IX (01) (22)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** Biology Paper–IX (01) (22)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. Cell wall is present in the cell of.

(A) Fungi only
(B) Plants only
(C) Plants and prokaryotes only
(D) All these

Q2. The longest part of the cell cycle is the.....stage.

(A) Interphase
(B) Prophase
(C) Metaphase
(D) Anaphase

Q3. Trypsin, an enzyme works best at a pH of.....

(A) 2
(B) 8
(C) 3
(D) 4

Q4. Light reactions takes place in the.....of chloroplast.

(A) Stroma
(B) Granum
(C) Cristae
(D) Cisternae

Q5. The first part of small intestine is called:

(A) Oesophagus
(B) Jejunum
(C) Duodenum
(D) Ileum

Q6. Red blood cells are also known as.....

(A) Leucocytes
(B) Lymphocytes
(C) Thrombocytes
(D) Erythrocytes

Q7. Branch of biology which deals with the study of composition, structure and functions of plant and animal tissues is called.....

(A) Morphology
(B) Anatomy
(C) Histology
(D) Physiology

Q8. The level of organization which is represented by the heart of frog is.

(A) Organ
(B) Tissue
(C) Organism
(D) Organelle

Q9. Which one of the given is measureable?

(A) Colour
(B) Smell
(C) Length
(D) Taste

Q10. Which kingdom includes organisms with cell wall but without chlorophyll?

(A) Protista
(B) Fungi
(C) Plantae
(D) Animalia

Q11. The zoological name of cat is.

(A) Felis Leo
(B) Felis Tigris
(C) Felis Domesticus
(D) Canis Lupus

Q12. The resolution of human eye is.

(A) 0.04 mm
(B) 0.1 mm
(C) 0.001 mm
(D) 1.0 mm

Section-B

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 32

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

QII. Answer any EIGHT parts. Each part carries FOUR marks.

1. Write down the linkage of Biology with Biochemistry and Biostatistics.
2. What are A.F.A King's observations about malaria.
3. Describe the contributions of Abu Usman Aljahiz in the field of Biology.
4. Differentiate between light and electron microscopes.

5. Define cell cycle. Write only names of events occurring during interphase.
6. How variation in temperature affects enzyme activity?
7. Compare lactic acid fermentation with alcoholic fermentation.
8. What are inorganic fertilizers? Describe their any three environmental hazards.
9. Describe liver as digestive gland.
10. Briefly describe arteriosclerosis. (Hardening of arteries)
11. Differentiate between transpiration and translocation.

Section-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 21

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [4] Differentiate between inductive reasoning and deductive reasoning.

(b) [3] Differentiate between two kingdom system and five kingdom system along with their advantages and disadvantages.

QIV.

(a) [3] Describe that Photosynthesis is a life sustaining process.

(b) [4] Describe apoptosis, give at least three examples.

QV.

(a) [4] Describe the effects of concentration of substrate and pH on enzyme activity.

(b) [3] Draw the structure of ATP molecule.

QVI.

(a) [4] Draw labeled diagram of human digestive system.

(b) [3] Describe the main functions of blood cells.